



CAREER POINT
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PIXSCOUT AI

AI-POWERED OFFLINE VISUAL SEARCH &
SECURE DOCUMENT ASSISTANT



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INTRODUCTION & BACKGROUND

INTRODUCTION

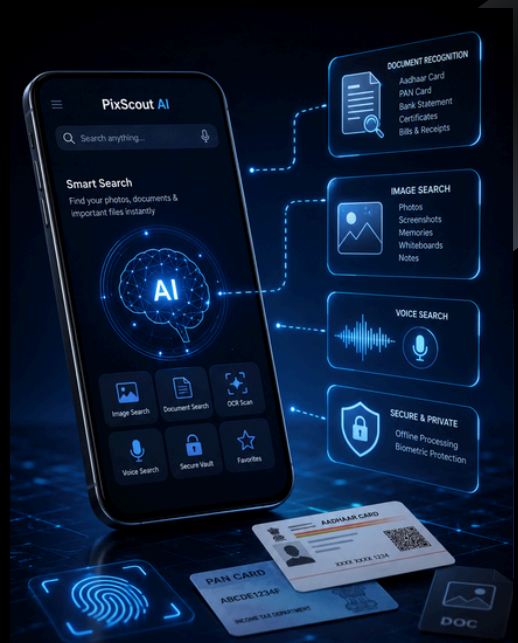
Smartphones have become an essential part of our daily lives, storing thousands of photos, screenshots, and important documents such as Aadhaar cards, PAN cards, bank statements, certificates, bills, and receipts. During emergencies, finding a specific file manually is often slow and frustrating because traditional gallery applications rely mainly on filenames, folders, or dates instead of understanding the actual content of images.

PixScout AI is an intelligent offline mobile application that solves this problem by combining Artificial Intelligence (AI), Computer Vision, Optical Character Recognition (OCR), and Natural Language Processing (NLP). The application allows users to search photos and documents using simple text or voice commands while keeping personal data secure through offline processing and biometric authentication. PixScout AI provides a faster, smarter, and privacy-focused solution for intelligent file management on Android devices.

BACKGROUND

With the rapid increase in smartphone storage capacity, users accumulate thousands of media files over time. Existing gallery applications provide only basic search functionality and cannot accurately identify images based on their visual content or extracted text. As a result, users waste valuable time searching for important documents when they are needed most.

PixScout AI addresses this challenge by providing a smart offline search system that understands image content, recognizes text inside documents, and retrieves the required file instantly. The proposed solution focuses on speed, privacy, security, and user convenience, making document management more efficient and reliable.





OBJECTIVES & SCOPE

THE OBJECTIVES OF THIS PROJECT ARE:

- ✕ Develop an AI-powered offline visual search application for Android devices.
- ✕ Enable intelligent image and document search using AI and Computer Vision.
- ✕ Extract text from images using Optical Character Recognition (OCR).
- ✕ Support both text-based and voice-based search for quick file retrieval.
- ✕ Ensure user privacy through offline processing and biometric authentication.

THE PROJECT WILL COVER THE FOLLOWING AREAS:

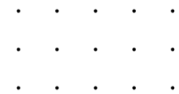
- ✕ AI-Based Image Recognition
- ✕ Optical Character Recognition (OCR)
- ✕ Natural Language Processing (NLP)
- ✕ Offline Processing & Privacy Protection
- ✕ Android Application Development (Flutter + FastAPI)





METHODOLOGY & TIMELINE

THE PROJECT WILL BE CARRIED OUT
USING THE FOLLOWING STEPS:



✕ Requirement Analysis

✕ AI Model Development

✕ Application Development

✕ Testing & Deployment

PROJECT TIMELINE

Phase	Activity	Duration
Phase 1	Requirement Analysis & Research	Week 1-2
Phase 2	UI Design & AI Module Development	Week 3-4
Phase 3	OCR, Voice Search & Backend Integration	Week 5-10
Phase 4	Testing, Documentation & Final Deployment	Week 11-12

PixScout AI follows a structured software development process. The project begins with requirement analysis and system design, followed by the development of AI-powered image recognition, OCR, voice search, and secure offline storage modules. After implementation, the application is thoroughly tested, optimized, and documented before deployment to ensure reliability, security, and an excellent user experience.





TECHNOLOGY STACK & CONCLUSION

PixScout AI is developed using modern Artificial Intelligence and mobile application technologies. The system combines Computer Vision, OCR, Natural Language Processing, and secure offline processing to provide fast, intelligent, and privacy-focused visual search for Android users.

Technology

Purpose

Flutter	Android Mobile Application
FastAPI	Backend API Development
TensorFlow Lite	On-device AI Inference
OpenCV	Image Processing
Tesseract OCR	Text Extraction from Images
SQLite	Offline Local Database

CONCLUSION

PixScout AI is an innovative offline visual search and secure document assistant designed to help users quickly find images and documents using Artificial Intelligence. The project focuses on speed, privacy, and offline accessibility, making it a reliable solution for everyday file management. Future enhancements may include multilingual support, cloud synchronization, and advanced AI-based image understanding.